SECURITY

Digital Forensics Basics Visual Guide

Collect evidence that holds up —
chain of custody to timeline analysis.

ANIMATED WALKTHROUGH
Step through pages to play the animation

STEP 1/6

Chain of Custody

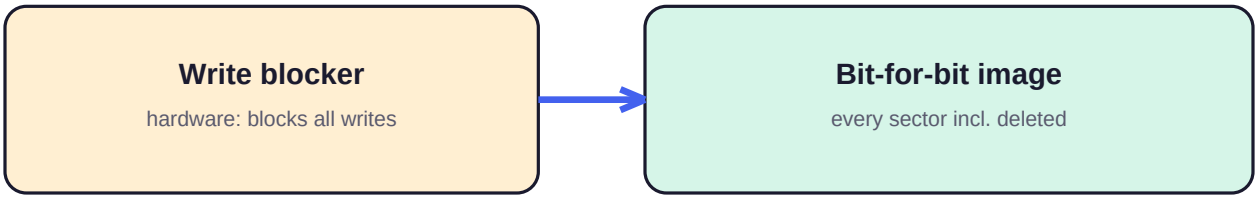
Evidence is worthless if you can't prove its journey.

Hash at the moment of capture; re-hash before analysis. Any gap in the log is reasonable doubt.

STEP 2/6

Disk Acquisition & Write Blockers

Copy everything, change nothing.



Analyze the COPY, never the original. Booting a suspect drive alters timestamps — the evidence starts degrading the moment it powers on.

STEP 3/6

Memory Acquisition

RAM holds what disk never sees.

- Capture memory FIRST — it vanishes on shutdown (running processes, keys, connections).
- Memory reveals: injected code, decrypted payloads, active C2 sessions.
- Use trusted capture tools from external media; each tool leaves its own footprint — document it.
- Order of volatility: memory → network → disk. Collect in that order.

STEP 4/6

Timeline Analysis

Reconstruct the story from timestamps.

Super-timeline: filesystem + logs + registry + browser, merged and sorted

one chronological view

- Anchor on known-good events, then walk outward to the suspicious.
- Correlate across sources: a file created at 02:14 matches the 02:14 firewall deny.
- Watch for timestomping — attackers forge timestamps; cross-check multiple sources.

STEP 5/6

Evidence Handling Rules

Discipline that survives cross-examination.

- Least privilege: only examiners touch originals; everyone else gets copies.
- Document every tool and command — reproducibility is credibility.
- Store originals sealed, hashed, and access-logged; encrypt in transit.
- Never work on live suspect systems longer than necessary to capture.

STEP 6/6

From Findings to Report

Facts first, opinions labeled.

- Report states WHAT was found and HOW (method, hashes) — not who to blame.
- Separate observed facts from analyst inferences, clearly marked.
- Include limitations: what wasn't captured, what couldn't be recovered.
- Preserve everything — cases reopen, and spoliation destroys credibility.

STEP ✓/✓

Key Takeaways

Forensics — Remember

- Chain of custody: seize, hash at capture, log every access.
- Write-block the original; analyze only bit-for-bit copies.
- Capture memory first — order of volatility: RAM, network, disk.
- Build super-timelines merging all sources; cross-check for timestomping.
- Document every tool and command; reproducibility is credibility.
- Report facts vs inferences separately; preserve everything.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev